

Software errors

No program is ever entirely error-free. Luckily, there are many techniques and tools that programmers can use to detect and fix these errors.

SEE ALSO

◀ 112–113 Assemblers, interpreters, and compilers

Language breakthroughs **122–123 ▶**

Maintenance and support **174–175 ▶**

Bugs in a program

An important part of being a programmer is the ability to recognize software errors, also called “bugs”, and fix them. There are three types of bugs – syntax, logic, and runtime errors. While some bugs cause obvious crashes and are easy to locate, others are subtle and can take months to find.

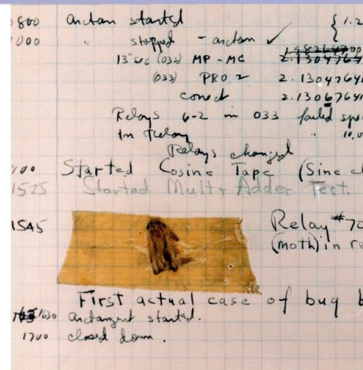
▽ Syntax errors

A syntax error is a typo or a small mistake, introduced by a programmer into the wording of the program. The compiler – which translates programming languages into machine code – will not work until all syntax errors are fixed.

IN DEPTH

First bug

In 1945, computers filled entire rooms and produced a lot of heat that attracted bugs, which crawled inside the machines and caused short circuits. On 9 September 1945, American computer scientist Grace Hopper (1906–1992) found that a moth had caused a malfunction in the Harvard Mark II computer, and she taped the moth into the computer’s log book. The term “bug” for a computer problem has been used ever since.



Typos and misspelled words

This includes missing brackets, semicolons, or quotation marks.

Undeclared variables

If a programmer forgets to declare a variable before assigning its value, the compiler won't be able to find the variable.

Off by one error

This includes overestimating the size of an array, or forgetting that arrays start at 0 and not 1.

Forgetting to link code

If the program is not written properly, the compiler won't be able to access libraries of pre-written code.

Integer division

Integer division truncates numbers by throwing away remainders, which can cause calculation mistakes if you need precision.

▷ Logic errors

A logic error is a flaw in the program's design that causes unanticipated behaviour. These bugs can be harder to find as they don't always produce crashes.